SOCIOL 723: Social Statistics II

Week 1 (Thursday), OLS in Matrix Form

Wenhao Jiang

Department of Sociology, Fall 2026

Duke University

Today

The Matrix Form and Its Geometry

Partialling Out in General

Tuesday built everything in scalar form. Today: the same objects again in matrix form, for general understanding, followed by lab.

★ = essential, you must understand this ★ = for understanding, not examined

Matrix Form

Why Bother

- With k regressors, scalar algebra needs k normal equations and becomes unreadable. Matrix algebra stays two lines long for any k.
- The geometry, projection onto a subspace, is invisible in scalar form and obvious in matrix form.
- In the literature, most estimators you will meet (fixed effects, IV, 2SLS, ridge) are *written* as small variations on $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$. We will keep working in scalars, but recognizing this form is what lets you read modern papers.

But

Nothing in this section is new content. It is the *same* normal equations you derived on Tuesday, written more compactly. And the lab stays entirely in scalar form: sums, means, and `lm()`, on simulations and the GSS.

Vectors and Matrices, From Zero ★

No linear algebra is assumed in this course. Here is the part we need, from nothing.

- A **vector** is a list of numbers written in a column. X_i is the list describing unit i: one entry per variable, with a 1 on top for the intercept.
- A **matrix** is a rectangular table of numbers: an $n \times k$ matrix has n rows and k columns.
- The **transpose** $'$ tips a column onto its side: if X_i is a $k \times 1$ column, then X_i' is the same list lying down, $1 \times k$.

The one formula to internalize

$$X_i'\boldsymbol{\beta} = \beta_0 \cdot 1 + \beta_1 X_{i1} + \cdots + \beta_{k-1} X_{i,k-1} :$$

a row times a column is a *single number*: multiply matching entries, then add. So $X_i'\boldsymbol{\beta}$ is nothing but unit i's fitted value, written compactly. Whenever you see $X_i'\boldsymbol{\beta}$, read it as "the regression line evaluated at unit i."

Multiplying and Inverting

The only two operations we use. **Multiplication** extends the row-times-column rule: entry (r, c) of $\boldsymbol{AB}$ is (row r of $\boldsymbol{A}$) times (column c of $\boldsymbol{B}$):

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}.$$

Dimensions must match, $(n \times k)(k \times 1) = (n \times 1)$, and order matters: $\boldsymbol{AB} \neq \boldsymbol{BA}$ in general.

Inversion is the matrix version of dividing. For a number, $a^{-1}a = 1$; for a square matrix, $\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}$, the *identity* matrix (ones on the diagonal, zeros elsewhere), which leaves whatever it multiplies unchanged.

- Just as $1/a$ fails to exist when $a = 0$, $\boldsymbol{A}^{-1}$ fails to exist when the matrix collapses information; that is the “rank” condition, explained shortly.
- You will never invert a matrix by hand here; `R`’s `solve()` does it. What matters is knowing what the symbol says.

Stacking the Data

For n units and k regressors (including the constant):

$$\boldsymbol{y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}_{n\times 1}, \quad \boldsymbol{X} = \begin{bmatrix} X_1' \\ \vdots \\ X_n' \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & \cdots & X_{1,k-1} \\ \vdots & \vdots & & \vdots \\ 1 & X_{n1} & \cdots & X_{n,k-1} \end{bmatrix}_{n\times k}, \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix}.$$

The model, stacked over units:

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \qquad \Longleftrightarrow \qquad Y_i = X_i'\boldsymbol{\beta} + \epsilon_i, \quad i = 1, \ldots, n.$$

Two objects do all the work: $\boldsymbol{X}'\boldsymbol{X} = \sum_i X_i X_i'$ (a $k \times k$ matrix) and $\boldsymbol{X}'\boldsymbol{y} = \sum_i X_i Y_i$ (a $k \times 1$ vector).

Least Squares in Matrix Form: the Expansion

The sum of squared errors is the error vector times itself:

$$S(b) = \sum_i (Y_i - X_i' b)^2 = (\boldsymbol{y} - \boldsymbol{X} b)'(\boldsymbol{y} - \boldsymbol{X} b).$$

Two transpose rules, used constantly

$(\boldsymbol{A} + \boldsymbol{B})' = \boldsymbol{A}' + \boldsymbol{B}'$, and $(\boldsymbol{AB})' = \boldsymbol{B}'\boldsymbol{A}'$: transposing a product *reverses the order*. Hence $(\boldsymbol{X} b)' = b'\boldsymbol{X}'$.

Multiply out, term by term:

$$S(b) = \boldsymbol{y}'\boldsymbol{y} - \boldsymbol{y}'\boldsymbol{X} b - b'\boldsymbol{X}'\boldsymbol{y} + b'\boldsymbol{X}'\boldsymbol{X} b = \boldsymbol{y}'\boldsymbol{y} - 2\, b'\boldsymbol{X}'\boldsymbol{y} + b'\boldsymbol{X}'\boldsymbol{X} b.$$

- Why do the middle terms merge? Each is 1×1, a single number, and a number equals its own transpose: $(b'\boldsymbol{X}'\boldsymbol{y})' = \boldsymbol{y}'\boldsymbol{X} b$. The two terms are the *same* number written twice.

Minimizing: Two Matrix Derivatives, From Zero

$\partial S/\partial b$ just stacks the k partials into a column; two stock rules cover everything:

The rules, with their scalar shadows

$$\frac{\partial(a'b)}{\partial b} = a \qquad \text{since } a'b = \textstyle\sum_j a_j b_j \text{ gives } \partial/\partial b_j = a_j \qquad (\text{cf. } \tfrac{d}{db}\, ab = a)$$

$$\frac{\partial(b'\boldsymbol{A}b)}{\partial b} = 2\boldsymbol{A}b \qquad \text{for symmetric } \boldsymbol{A} \qquad (\text{cf. } \tfrac{d}{db}\, ab^2 = 2ab)$$

Apply them with $a = \boldsymbol{X}'\boldsymbol{y}$ and $\boldsymbol{A} = \boldsymbol{X}'\boldsymbol{X}$ (symmetric):

$$\frac{\partial S(b)}{\partial b} = -2\boldsymbol{X}'\boldsymbol{y} + 2\boldsymbol{X}'\boldsymbol{X}b \overset{!}{=} 0 \quad \Longrightarrow \quad \underbrace{\boldsymbol{X}'\boldsymbol{X}\,\hat{\beta} = \boldsymbol{X}'\boldsymbol{y}}_{\text{normal equations}}.$$

The *same* k equations you wrote out one at a time on Tuesday. Why a minimum? $v'\boldsymbol{X}'\boldsymbol{X}v = \|\boldsymbol{X}v\|^2 \geq 0$ for all v: S is a bowl. (Shorthand: $\boldsymbol{X}'\boldsymbol{X} \succeq 0$, the matrix $\geq$.)

The Normal Equations, Slowly: the Bivariate Case

Take $k = 2$ with $X_i = (1, X_i)'$, so $b = (b_0, b_1)'$. The two building blocks are

$$\boldsymbol{X}'\boldsymbol{X} = \begin{bmatrix} n & \sum_i X_i \\ \sum_i X_i & \sum_i X_i^2 \end{bmatrix}, \qquad \boldsymbol{X}'\boldsymbol{y} = \begin{bmatrix} \sum_i Y_i \\ \sum_i X_i Y_i \end{bmatrix}.$$

Write out $\boldsymbol{X}'\boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{X}'\boldsymbol{y}$ one row at a time:

$$n\hat\beta_0 + \hat\beta_1 \sum_i X_i = \sum_i Y_i \iff \sum_i \left(Y_i - \hat\beta_0 - \hat\beta_1 X_i\right) = 0$$

$$\hat\beta_0 \sum_i X_i + \hat\beta_1 \sum_i X_i^2 = \sum_i X_i Y_i \iff \sum_i X_i\left(Y_i - \hat\beta_0 - \hat\beta_1 X_i\right) = 0$$

- Row by row, the matrix equation *is* Tuesday's (N1) and (N2): residuals sum to zero, and residuals are orthogonal to the regressor.
- One matrix line, k scalar equations. That is the entire content of the compact notation: nothing new, everything stacked.

The Estimator, Existence, and Rank

What "rank" means, in words

The rank of $\boldsymbol{X}$ is the number of genuinely distinct columns it contains: columns that cannot be rebuilt as combinations of the others. *Full column rank*, $\text{rank}(\boldsymbol{X}) = k$, says no column is redundant: the data carry k separate pieces of information, one per coefficient.

The OLS estimator

If $\text{rank}(\boldsymbol{X}) = k$, $\boldsymbol{X}'\boldsymbol{X}$ is invertible and

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} = \left(\mathbb{E}_n[X_i X_i']\right)^{-1}\mathbb{E}_n[X_i Y_i].$$

- Full rank needs $n \geq k$; it is the matrix version of "X must vary."
- Failures you will actually meet: the *dummy variable trap*; including a variable and a rescaling of it; a group indicator constant within a fixed effect.
- If rank fails, `R` silently drops a column and reports `NA`. That `NA` is never a mystery.

Sanity Check: Recovering the Scalar Formula ★

Let $k = 2$ with $X_i = (1, X_i)'$. Then

$$\boldsymbol{X}'\boldsymbol{X} = \begin{bmatrix} n & \sum_i X_i \\ \sum_i X_i & \sum_i X_i^2 \end{bmatrix}, \qquad \boldsymbol{X}'\boldsymbol{y} = \begin{bmatrix} \sum_i Y_i \\ \sum_i X_i Y_i \end{bmatrix}.$$

Using $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$,

$$\hat{\beta}_1 = \frac{n\sum_i X_i Y_i - \sum_i X_i \sum_i Y_i}{n\sum_i X_i^2 - (\sum_i X_i)^2} = \frac{\mathrm{Cov}_n(X, Y)}{\mathrm{Var}_n(X)}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

Exactly what we derived by hand. The matrix form is a change of language, not of substance.

Why Projection? It Is Tuesday's Argmin, Re-Read ★

Tuesday defined OLS by a minimization: $\hat{\beta} = \arg\min_b \sum_{i=1}^n (Y_i - X_i'b)^2$.

The one observation everything hangs on

A sum of squared entries is a squared *length*: $\sum_i (Y_i - X_i'b)^2 = \|\boldsymbol{y} - \boldsymbol{X}b\|^2$, the squared *distance* between two points of $\mathbb{R}^n$, the data $\boldsymbol{y}$ and the candidate fit $\boldsymbol{X}b$. So the argmin says, word for word: among all fits the model can produce, pick the one closest to the data.

- The candidate fits $\boldsymbol{X}b$ form a flat sheet, a "plane" (next frame), and "closest point on a plane" is a problem geometry solved long before calculus: the foot of the perpendicular.
- The right angle *is* the first-order condition: perpendicular to every column of $\boldsymbol{X}$ means $\boldsymbol{X}'\hat{\boldsymbol{e}} = \boldsymbol{0}$, the normal equations, with no differentiation anywhere.
- Nothing new is computed: projection is the same argmin, *answered by geometry instead of calculus*. Next: the smallest possible case.

What Does "All the Fits" Look Like? ★

Read matrix-times-vector as a recipe for mixing columns:

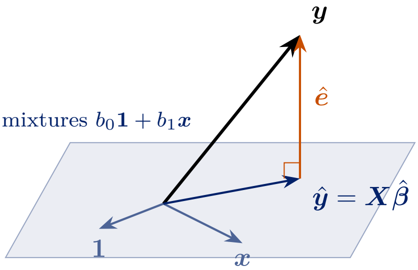
$$\boldsymbol{X}b = b_1 \cdot (\text{column } 1) + b_2 \cdot (\text{column } 2) + \cdots + b_k \cdot (\text{column } k) :$$

each coefficient pours in that much of its column. The set of candidate fits is therefore *every possible mixture of the k columns*.

- **One column** (intercept only): the mixtures are all multiples of $\mathbf{1}$, a line through the origin. Projecting $\boldsymbol{y}$ onto that line gives $\hat{b} = \bar{Y}$: the sample mean is a projection.
- **Two columns**: the mixtures of two arrows fill a flat plane through the origin.
- k **columns**: the mixtures fill a k-dimensional flat subspace of $\mathbb{R}^n$. "k-dimensional" simply counts the free coefficients $b_1, \ldots, b_k$.
- $\boldsymbol{y}$ generally sits *off* this subspace; the gap is the residual. A column that is itself a mixture of the others adds no new direction, so the subspace has dimension below k: rank, again.

The Picture: $n = 3$, Two Columns ★

Three observations ($n = 3$), two columns: $\boldsymbol{X} = [\,\mathbf{1} \;\; \boldsymbol{x}\,]$ is 3×2. The mixtures $b_0\mathbf{1} + b_1\boldsymbol{x}$ fill the shaded plane; the data $\boldsymbol{y}$ floats *off* it, in the third dimension.



- OLS drops the perpendicular from $\boldsymbol{y}$ to the plane: the foot is the fit $\hat{\boldsymbol{y}}$, the gap is the residual $\hat{\boldsymbol{e}}$. Perpendicular to the plane = perpendicular to *both* columns: $\boldsymbol{X}'\hat{\boldsymbol{e}} = \mathbf{0}$, the normal equations.
- Fitting is dropping a perpendicular. Everything else is bookkeeping.

Deriving $\hat{\beta}$ from the Picture, No Calculus ★

The picture makes its own demand: the residual must be perpendicular to the plane, that is, perpendicular to *every column* of $\boldsymbol{X}$. Written in one matrix line, and then rearranged:

$$\boldsymbol{X}'(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}) = \mathbf{0} \quad \Longrightarrow \quad \boldsymbol{X}'\boldsymbol{X}\,\hat{\boldsymbol{\beta}} = \boldsymbol{X}'\boldsymbol{y} \quad \Longrightarrow \quad \hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}.$$

- The first equality *is* the perpendicularity requirement; the rest is rearranging. Same normal equations, same estimator, and not a derivative in sight. Geometry and calculus are two proofs of one formula.
- Substituting back, the fitted vector is $\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$: one matrix carries the data straight to the fit.

Names you will meet in the literature: $\boldsymbol{P} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$ is the *hat matrix* (it puts the hat on $\boldsymbol{y}$), and $\boldsymbol{M} = \boldsymbol{I}_n - \boldsymbol{P}$ the *residual maker* ($\hat{\boldsymbol{e}} = \boldsymbol{M}\boldsymbol{y}$). We will not need their algebra.

Partialling Out

The Frisch–Waugh–Lovell Theorem ★

Partition $\boldsymbol{X} = [\,\boldsymbol{X}_1 \;\; \boldsymbol{X}_2\,]$: a treatment of interest and controls. Let $\boldsymbol{M}_2 = \boldsymbol{I} - \boldsymbol{X}_2(\boldsymbol{X}_2'\boldsymbol{X}_2)^{-1}\boldsymbol{X}_2'$ and define

$$\tilde{\boldsymbol{X}}_1 = \boldsymbol{M}_2\boldsymbol{X}_1, \qquad \tilde{\boldsymbol{y}} = \boldsymbol{M}_2\boldsymbol{y}.$$

Theorem

The OLS estimate $\hat{\boldsymbol{\beta}}_1$ from the full regression equals

$$\hat{\boldsymbol{\beta}}_1 = (\tilde{\boldsymbol{X}}_1'\tilde{\boldsymbol{X}}_1)^{-1}\tilde{\boldsymbol{X}}_1'\,\tilde{\boldsymbol{y}} \;=\; (\tilde{\boldsymbol{X}}_1'\tilde{\boldsymbol{X}}_1)^{-1}\tilde{\boldsymbol{X}}_1'\,\boldsymbol{y} \qquad \text{(note: } \tilde{\boldsymbol{y}} \text{ vs. raw } \boldsymbol{y}\text{)},$$

The two right-hand sides differ only in the tilde on $\boldsymbol{y}$; they are equal for Tuesday's reason (the part removed from $\boldsymbol{y}$ is built from $\boldsymbol{X}_2$, to which $\tilde{\boldsymbol{X}}_1$ is orthogonal). Tuesday's three-step recipe, any number of controls.

Proof of FWL ★

Start from the *fitted* full regression, an exact identity in the sample:

$$\boldsymbol{y} = \boldsymbol{X}_1\hat{\boldsymbol{\beta}}_1 + \boldsymbol{X}_2\hat{\boldsymbol{\beta}}_2 + \hat{\boldsymbol{e}}.$$

Premultiply once by $\boldsymbol{M}_2$, using $\boldsymbol{M}_2\boldsymbol{X}_2 = \boldsymbol{0}$ and $\boldsymbol{M}_2\hat{\boldsymbol{e}} = \hat{\boldsymbol{e}}$ ($\hat{\boldsymbol{e}}$ already has $\boldsymbol{X}_2$ removed, since $\boldsymbol{X}_2'\hat{\boldsymbol{e}} = \boldsymbol{0}$):

$$\tilde{\boldsymbol{y}} = \tilde{\boldsymbol{X}}_1\hat{\boldsymbol{\beta}}_1 + \hat{\boldsymbol{e}}.$$

Now regress $\tilde{\boldsymbol{y}}$ on $\tilde{\boldsymbol{X}}_1$:

$$(\tilde{\boldsymbol{X}}_1'\tilde{\boldsymbol{X}}_1)^{-1}\tilde{\boldsymbol{X}}_1'\tilde{\boldsymbol{y}} = \hat{\boldsymbol{\beta}}_1 + (\tilde{\boldsymbol{X}}_1'\tilde{\boldsymbol{X}}_1)^{-1}\underbrace{\tilde{\boldsymbol{X}}_1'\hat{\boldsymbol{e}}}_{=\boldsymbol{0}} = \hat{\boldsymbol{\beta}}_1,$$

because $\tilde{\boldsymbol{X}}_1'\hat{\boldsymbol{e}} = \boldsymbol{X}_1'\boldsymbol{M}_2\hat{\boldsymbol{e}} = \boldsymbol{X}_1'\hat{\boldsymbol{e}} = \boldsymbol{0}$ (using $\boldsymbol{M}_2' = \boldsymbol{M}_2$, visible by transposing its definition). ■

Why FWL Matters More Than It Looks

1. **Interpretation.** "Controlling for W" = discarding all covariance between X_1 and W. Your estimate lives on what remains.
2. **Diagnostics.** The added-variable plot is the only two-dimensional picture that faithfully represents a multivariate slope.
3. **Generality.** Nothing in the argument requires $\boldsymbol{X}_2$ to enter linearly, it only requires that we can compute the part of X_1 that $\boldsymbol{X}_2$ predicts. That observation is the seed of a large literature, and we return to it briefly in Week 13.

Where This Leaves Us

- OLS minimizes squared errors. Everything else, the slope formula, the normal equations, FWL, OVB, follows from that one criterion by algebra.
- In the population, OLS estimates the best linear approximation to the CEF. Under A1–A4 the sample version is unbiased for it.
- Whether that coefficient is a *causal effect* is a separate question that regression cannot answer.
- **Next week**: we have $\hat{\beta}$; now we need to know how uncertain it is, and the standard error your software prints is usually the wrong one.

This week's reading

MHE Chapter 3; CCI Chapter 6. The lab computes OLS by hand in scalar form, on simulated data and the GSS; no matrix computations anywhere.

References

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